

```

}
void salgo (no,z,uu,u)

/* Consider simplest algorithm for the Sturm-Liouville equation.*/

int no;
double z,uu;
double u[];
{
int i;
double z2,q,p,p1,x;
q = uu*(1+uu);
z2 = 2*z*x;
for (i = 1; i < no-1; ++i)
{
p1 = -.2*x*x;
p = 2*(1-x*x);
x = -1+i*x;
u[i+1] = ((2*p-z2*q)*u[i]+(p1-p)*u[i-1])/(p1+p);
}
}

```

You can extend this algorithmic method to solve other problems as many equations like the Bessel equation, which is given by...

$$x^2 y'' + xy' + (\lambda^2 x^2 - \nu^2)y = 0$$

...can be represented in the Sturm-Liouville form as

$$(xy') + (\lambda^2 x^2 - \nu^2/x)y = 0$$

We will now try out Lagrange interpolation using the Aitken method. I'm sure you must have done a simple form of this already. But this one is slightly diverse.

```

/* Main program for the Lagrange interpolation with the Aitken method.*/

#include <stdio.h>
#include <math.h>
#define NOMAX 3

main()

{
int i;
double f, dfi;
double h = 0.5, x = 0.9;
double xi[NOMAX], fi[] = {1.1, 0.9, 0.7, 0.5, 0.3};
void aitken_method_kernel_voyage();
for (i = 0; i < NOMAX; ++i)
{
xi[i] = h*i;
}
aitken_method_kernel_voyage(NOMAX, xi, fi, x, &f, &dfi);

```

To my readers

Thanks for your response, feedback and suggestions. But I am extremely sorry that I am not in a position to reply to all the queries made through e-mails. The major problem is that e-mails often get messed up. Hence, I may miss your e-mail. Sometimes I mark your mails, but the new e-mails often mask them. To avoid this problem, I have created a new platform for answering questions:

aasisvinayak.com/new_zone/forum.php

Considering the suggestions from readers, the next column will focus on notations and numerical analysis. But the emphasis will be on the designing of algorithms, as this is a kernel-programming column and not a section that addresses pure scientific computing.

```

printf("%10.6f %10.6f %10.6f\n", x, f, dfi);
}

void aitken_method_kernel_voyage(n, xi, fi, x, f, dfi)

int n;
double xi[], fi[];
double x;
double *f, *dfi;
{
int i, j;
double fn(NOMAX);
double x1, x2, f1, f2;
for (i = 0; i <= (n-1); ++i)
{
fn[i] = fi[i];
}
for (i = 0; i <= (n-2); ++i)
{
for (j = 0; j <= (n-i-2); ++j)
{
x1 = xi[j];
x2 = xi[j+i+1];
f1 = fn[j];
f2 = fn[j+i];
fn[j] = (x-x1)/(x2-x1)*f2+(x-x2)/(x1-x2)*f1;
}
}
*f = fn[0];
*dfi = (fabs(*f-f1)+fabs(*f-f2))/2;
}

```

You can see the output by executing this code in your compiler. (You can try changing the values assigned and see what happens.)

Iterations in coding

As we discussed before, iteration plays an important role when it comes to designing algorithms. Now, we will try to find the value of Pi using this.

```

/* Estimating Pi - Iteration*/

```